

Supplementary Context-Level Robustness

The main paper reports aggregate performance and retains the context-level analysis as a robustness check. Within each dataset-backbone context, the non-proposed reference method was selected by validation macro-F1 before the frozen test comparison. Table S1 summarizes absolute macro-F1 by dataset, while Table S2 separates numerical, practical, and paired statistical interpretations.

TABLE S1: Dataset-level comparison against context-wise validation-selected non-proposed reference methods.

Dataset	C	n	HMDF-kNN	Val.-selected ref.	Delta	Most frequent ref.	Reference methods selected	Practical W/T/L	Stat. W/NS/L (n)
Brain MRI 15C	15	9	0.9810	0.9744	+0.0067	MAXVAR-GCCA GCCA/MvDA (3)	MAXVAR-GCCA (3); MvDA (3); NCA+kNN (2); Final emb. (1)	6/3/0	1/7/0 (8)
Brain MRI 17C	17	9	0.9842	0.9821	+0.0021	MAXVAR-GCCA (7)	MAXVAR-GCCA (7); MvDA (1); NCA+kNN (1)	4/4/1	0/9/0 (9)
Brain MRI 44C	44	9	0.9428	0.9414	+0.0014	MAXVAR-GCCA (6)	MAXVAR-GCCA (6); Final emb. (1); NCA+kNN (1); MvDA (1)	2/6/1	0/8/0 (8)
Brain MRI 4C	4	9	0.9497	0.9498	-0.0002	Final emb./GMLDA (3)	Final emb. (3); GMLDA (3); EasyMKL (1); MLCFF (1); Softmax (1)	0/9/0	0/6/0 (6)
ScienceDB 3C	3	9	0.9504	0.9311	+0.0192	MAXVAR-GCCA (5)	MAXVAR-GCCA (5); Final emb. (2); MvDA (1); EasyMKL (1)	7/2/0	1/6/0 (7)

Within each dataset-backbone context, the non-proposed reference is selected by validation macro-F1 before test comparison; the displayed reference macro-F1 is the mean of those context-wise selections. Practical W/T/L uses a 0.005 margin. Statistical W/NS/L uses only contexts with paired predictions and applies Holm correction.

TABLE S2: Context-level outcomes against validation-selected reference methods.

Criterion	n	HMDF ahead	Tie/NS	Ref. ahead
Numerical sign	45	28	0	17
Practical margin (0.005)	45	19	24	2
Paired bootstrap 95% CI	38	10	28	0
Paired bootstrap + Holm	38	2	36	0

Numerical and practical outcomes use all 45 contexts. NS denotes not statistically significant among contexts with paired predictions. Practical ties are defined only by the 0.005 margin and are not equivalent to $p > 0.05$. Holm correction controls family-wise error over the available paired tests.

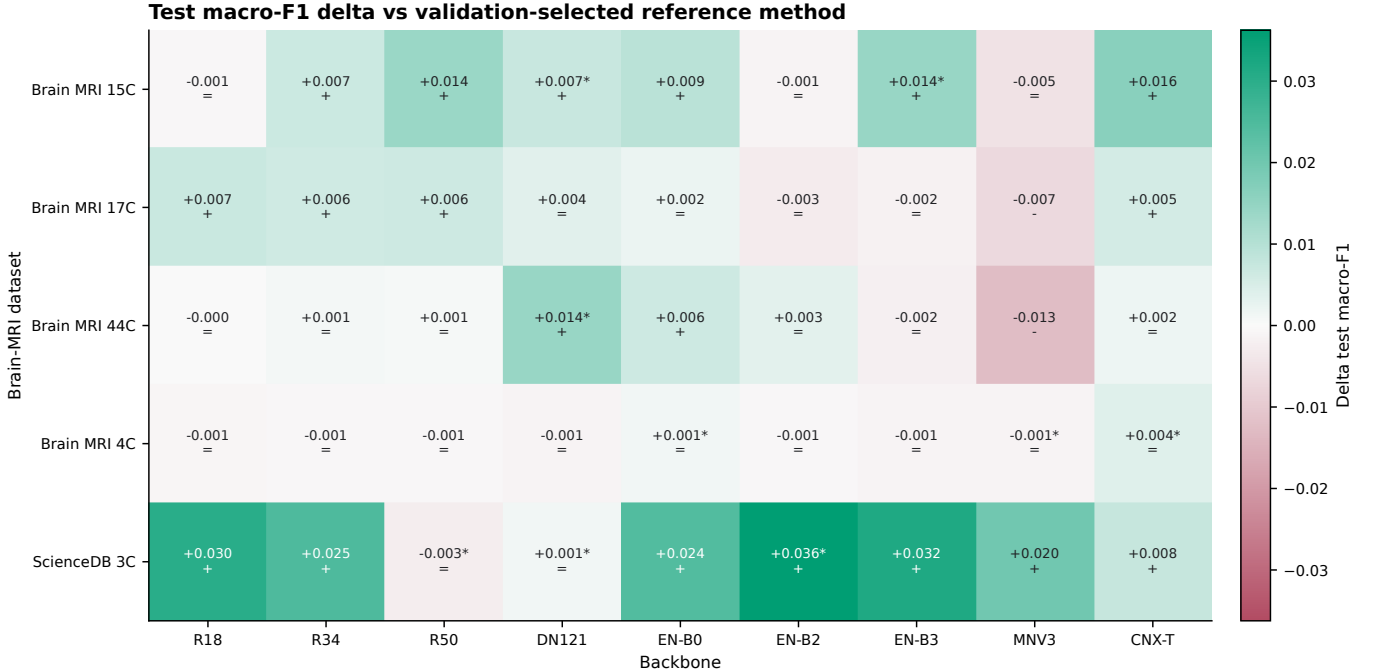


Fig. S1: Context-level test macro-F1 delta between HMDF-kNN and the strongest validation-selected non-proposed reference method in the same dataset-backbone context. Positive values favor HMDF-kNN. Symbols indicate outcomes using a 0.005 practical margin. An asterisk marks a paired bootstrap difference that remains significant after Holm correction among contexts with aligned prediction artifacts.

Holm-significant practical win: ScienceDB 3C (EN-B2)

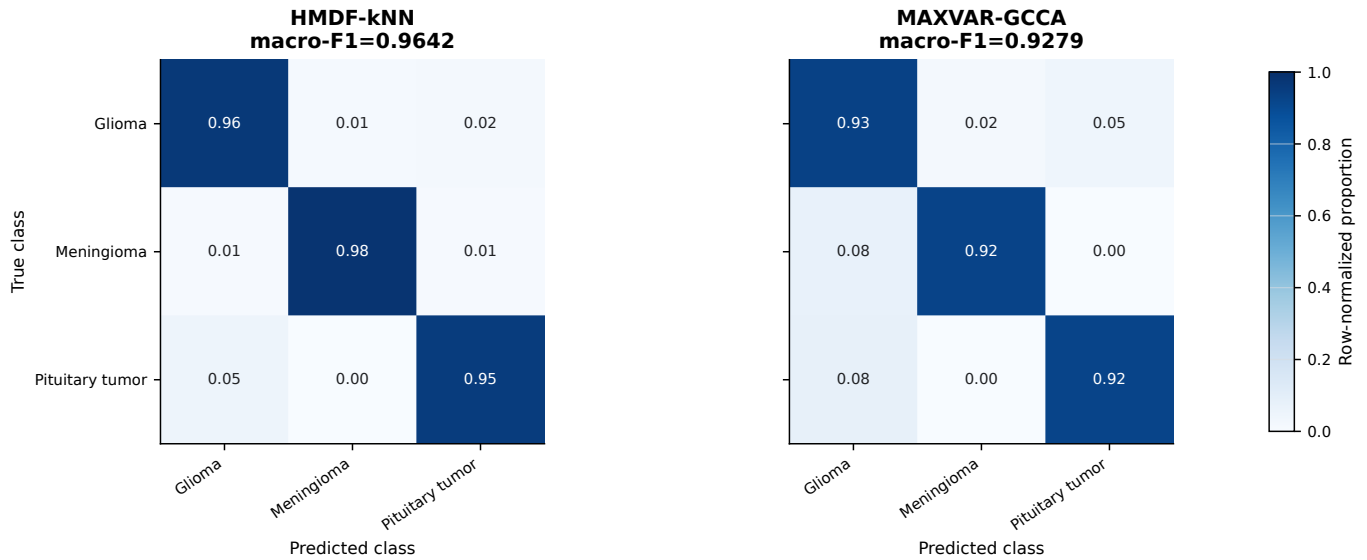


Fig. S2: Row-normalized confusion matrices for the ScienceDB 3C EfficientNet-B2 context used for class-level inspection. HMDf-kNN and MAXVAR-GCCA were evaluated on the same frozen test samples. The context was selected as the largest macro-F1 difference among the Holm-corrected significant practical gains and is shown as an error-analysis example, not as clinical validation.

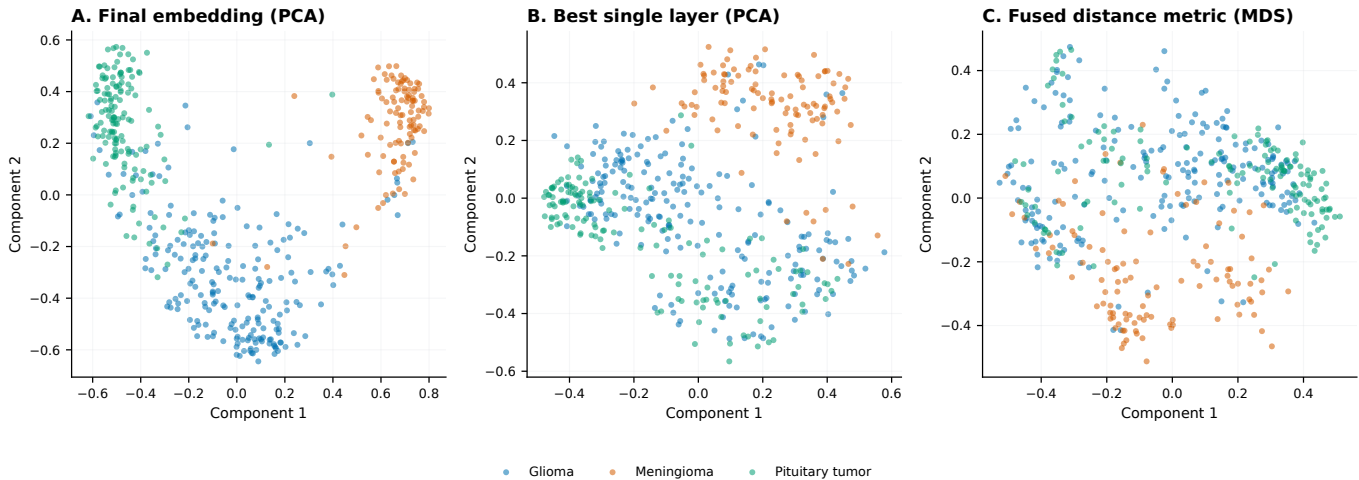


Fig. S3: Qualitative geometry for the same ScienceDB 3C EfficientNet-B2 context. The final embedding and best validation-ranked single layer are projected with PCA. The fused panel applies classical multidimensional scaling to the validation-selected weighted distance matrix. This visualization is not used for model selection or quantitative evaluation.